

Basic Formula

For an input vector

$$X = [x_1, x_2]$$

the perceptron computes:

$$net = w_1x_1 + w_2x_2 + b$$

Using the step activation:

$$y = \begin{cases} 1, & net \geq 0 \\ 0, & net < 0 \end{cases}$$

Error:

$$e = t - y$$

Weight update:

$$w_i^{new} = w_i^{old} + \eta ex_i$$

Bias update:

$$b^{new} = b^{old} + \eta e$$

where:

- t = target
- y = predicted output
- η = learning rate

Implement the OR gate using the perceptron learning algorithm.

Use:

$$\eta = 0.1$$

and initialize:

$$w_1 = w_2 = b = 0$$

Training data:

x_1	x_2	Target
0	0	0
0	1	1
1	0	1
1	1	1

Input 1

$$X = (0, 0), \quad t = 0$$

Initial:

$$w_1 = 0, w_2 = 0, b = 0$$

$$net = 0$$

Therefore:

$$y = 1$$

Error:

$$e = 0 - 1 = -1$$

Update:

$$w_1 = 0 + 0.1(-1)(0) = 0$$

$$w_2 = 0 + 0.1(-1)(0) = 0$$

$$b = 0 + 0.1(-1) = -0.1$$

Input 2

$$X = (0, 1), \quad t = 1$$

$$net = (0)(0) + (0)(1) - 0.1 = -0.1$$

Therefore:

$$y = 0$$

$$e = 1 - 0 = 1$$

Update:

$$w_1 = 0$$

$$w_2 = 0 + 0.1(1)(1) = 0.1$$

$$b = -0.1 + 0.1 = 0$$

Input 3

$$X = (1, 0), \quad t = 1$$

$$net = (0)(1) + (0.1)(0) + 0 = 0$$

$$y = 1$$

$$e = 0$$

No update.

Input 4

$$X = (1, 1), \quad t = 1$$

$$net = 0 + 0.1 + 0 = 0.1$$

$$y = 1$$

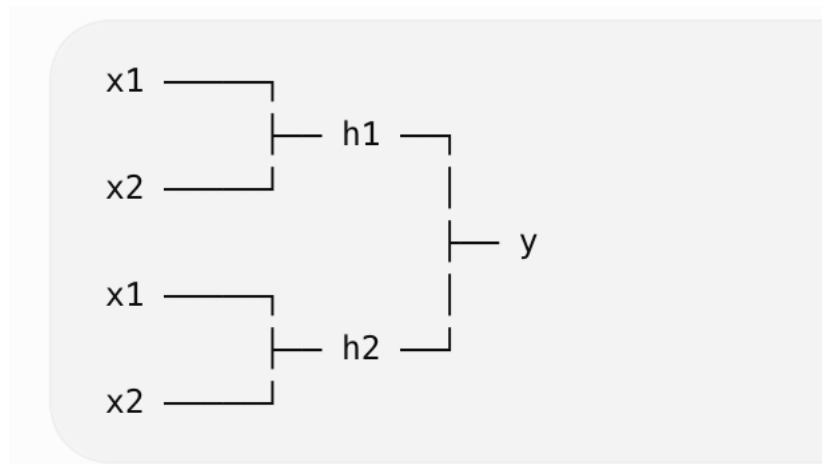
$$e = 0$$

End of first epoch

$w_1 = 0, \quad w_2 = 0.1, \quad b = 0$

Further epochs continue until all patterns are classified correctly.

Consider a two layer network



Given:

$$x_1 = 1, \quad x_2 = 0$$

Hidden-layer weights:

$$w_{11} = 0.5, \quad w_{21} = 0.4$$

$$w_{12} = 0.3, \quad w_{22} = 0.7$$

Hidden biases:

$$b_1 = 0.1, \quad b_2 = 0.2$$

Output weights:

$$v_1 = 0.6, \quad v_2 = 0.8$$

Output bias:

$$b_o = 0.1$$

Use sigmoid activation throughout.

Find the output.

Solution

Hidden neuron 1

$$z_1 = (1)(0.5) + (0)(0.4) + 0.1$$

$$z_1 = 0.6$$

$$h_1 = \sigma(0.6)$$

$$h_1 \approx 0.646$$

Hidden neuron 2

$$z_2 = (1)(0.3) + (0)(0.7) + 0.2$$

$$z_2 = 0.5$$

$$h_2 \approx 0.622$$

Output neuron

$$z_o = (0.646)(0.6) + (0.622)(0.8) + 0.1$$

$$z_o \approx 1.110$$

Therefore:

$$y = \sigma(1.110)$$

$$y \approx 0.752$$

For the above problem assume target $t=1$, finding the backward propagation

$$\delta_o = (t - y)y(1 - y)$$

Using:

$$y = 0.752$$

we get:

$$\delta_o = (1 - 0.752)(0.752)(1 - 0.752)$$

$$\delta_o \approx 0.0461$$

Thus:

$$\delta_o \approx 0.0461$$

Using:

$$\eta = 0.1$$

calculate the new output weights.

Solution

For v_1 :

$$\begin{aligned}\Delta v_1 &= \eta \delta_o h_1 \\ &= (0.1)(0.0461)(0.646) \\ \Delta v_1 &\approx 0.00298\end{aligned}$$

$$v_1^{new} = 0.6 + 0.00298$$

$$v_1^{new} \approx 0.603$$

For v_2 :

$$\begin{aligned}\Delta v_2 &= (0.1)(0.0461)(0.622) \\ \Delta v_2 &\approx 0.00287\end{aligned}$$

Therefore:

$$v_2^{new} \approx 0.803$$

Bias:

$$\begin{aligned}b_o^{new} &= b_o + \eta \delta_o \\ &= 0.1 + (0.1)(0.0461)\end{aligned}$$

$$b_o^{new} \approx 0.1046$$

The hidden-layer error signal is:

$$\delta_{h1} = h_1(1 - h_1)\delta_o v_1$$

Substituting:

$$\delta_{h1} = (0.646)(1 - 0.646)(0.0461)(0.6)$$

$$\delta_{h1} \approx 0.00634$$

For hidden neuron 2:

$$\delta_{h2} = h_2(1 - h_2)\delta_o v_2$$

$$\delta_{h2} \approx 0.00814$$

Using:

$$\eta = 0.1$$

calculate the updated weight from x_1 to hidden neuron 1.

Solution

$$\Delta w_{11} = \eta \delta_{h1} x_1$$

Since:

$$x_1 = 1$$

$$\Delta w_{11} = (0.1)(0.00634)(1)$$

$$= 0.000634$$

Therefore:

$$w_{11}^{new} = 0.5 + 0.000634$$

$$w_{11}^{new} = 0.500634$$

For $x_2 = 0$:

$$\Delta w_{21} = 0$$

Therefore:

$$w_{21}^{new} = 0.4$$

For a neural network:

$$2 \rightarrow 2 \rightarrow 1$$

perform:

1. Forward propagation
2. Calculate output error
3. Backpropagate error
4. Update output weights
5. Update hidden weights

using one training example.

The sequence is:

$$X \rightarrow Z^{(1)} \rightarrow A^{(1)} \rightarrow Z^{(2)} \rightarrow \hat{Y}$$

Then:

$$E = \frac{1}{2}(Y - \hat{Y})^2$$

Output error:

$$\delta^{(2)} = (Y - \hat{Y})\hat{Y}(1 - \hat{Y})$$

Output-layer gradient:

$$\frac{\partial E}{\partial W^{(2)}} = \delta^{(2)}(A^{(1)})^T$$

Hidden-layer error:

$$\delta^{(1)} = (W^{(2)})^T \delta^{(2)} \odot A^{(1)}(1 - A^{(1)})$$

Hidden-layer gradient:

$$\frac{\partial E}{\partial W^{(1)}} = \delta^{(1)} X^T$$

Parameter update:

$$W \leftarrow W - \eta \frac{\partial E}{\partial W}$$

This process is repeated for every training example and every epoch.